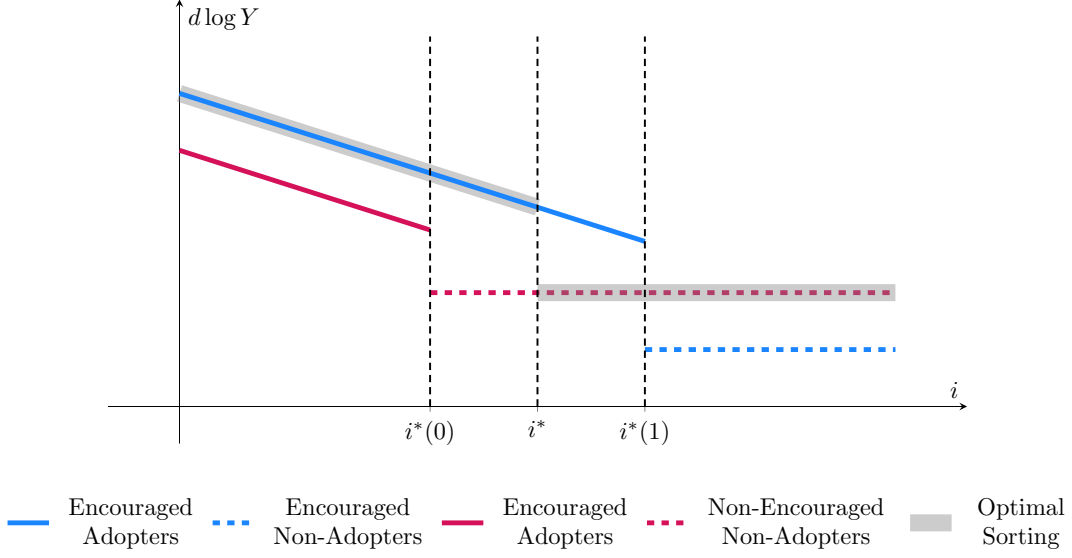


Figure A.2: The Impact of AI Chatbots on Worker Sorting, Adoption, and Earnings



Notes: This figure illustrates how AI chatbots affect workers' sorting, adoption, and earnings across the distribution of AI advantage. The x-axis ranks workers by chatbot advantage i ; the y-axis shows earnings impacts. Blue and red lines represent earnings effects for each worker-firm type combination. The gray line highlights equilibrium sorting and adoption patterns.

Workplace employment. The effect of encouragement on headcount employment (N) depends on whether increased human capital demand is offset by adoption-driven productivity gains. The increase in human capital demand is:

$$\log H_1 - \log H_0 = \frac{\epsilon}{\epsilon - 1} \kappa, \quad (30)$$

and the average productivity gain among encouraged adopters is:

$$\mathbb{E}[B_{ij} \mid i \leq i^*, E_j = 1] = (1 - \gamma)(\alpha_{b0} + \alpha_{b1}) + \gamma(\alpha_{c0} + \alpha_{c1} + \kappa/(\epsilon - 1)), \quad (31)$$

where $\gamma = \frac{\beta_b}{2\beta} \geq 0$, generalizing Equation (9) to account for worker sorting. Combining these, the employment effect is:

$$\log N_1 - \log N_0 = \frac{\epsilon - \gamma}{\epsilon - 1} \kappa - (1 - \gamma)(\alpha_{b0} + \alpha_{b1}) - \gamma(\alpha_{c0} + \alpha_{c1}). \quad (32)$$

Encouragement tends to *reduce* employment when encouragement is cheap (κ low), demand is elastic (ϵ high), workers are homogeneous in chatbot productivity (γ low), productivity

gains are large (α_b high), and adoption costs are high (α_c high).

Workplace wage bills. Encouraged firms increase wage bills by:

$$\log(W_1 H_1) - \log(W_0 H_0) = \kappa. \quad (33)$$

That encouragement unambiguously raises wage bills is a consequence of modeling chatbots solely as tools augmenting worker productivity (Equation (14)); as discussed in Section A.2.1, this need not hold under more general production technologies.

A.2.5 Empirical Identification

This section shows that causal effects of chatbots on worker and workplace outcomes can be identified using difference-in-differences, and provides structural interpretations of the estimands.

The derivations above assumed that differences across workers and firms arise solely from chatbot use and encouragement. More realistically, workers differ in baseline productivity ($H_{ij}(A_{ij}) = H_{i0} \exp(B_{ij} A_{ij})$) and firms in baseline demand ($Y_j = Y_{j0} P_j^{-\epsilon}$). For example, Figure 5(a) shows that adopters already earned more before chatbots became available.

Causal effects are still identified if baseline heterogeneity is time-invariant and log-additive, so that it differences out:

$$\Delta H_{ij} = B_{ij}(A_{ij}^{post} - A_{ij}^{pre}), \quad \Delta Y_j = -\epsilon(\log P_j^{post} - \log P_j^{pre}), \quad (34)$$

where $\Delta X = \log X^{post} - \log X^{pre}$. If baseline heterogeneity varies over time, identification requires that adoption and encouragement decisions are uncorrelated with anticipated shocks to H_{i0} and Y_{j0} —consistent with the policy rules in Equations (17) and (20).

Worker-level difference-in-differences. Equation (25) motivates three estimators used in Section F.2:

- 1. Productivity effect among non-encouraged adopters** (*Non-encouraged adopters vs. non-encouraged non-adopters*)

$$\Delta_{\text{NE,A} - \text{NE,NA}}^Y = \gamma\alpha_{c0} + (1 - \gamma)\alpha_{b0}, \quad (35)$$

where $\gamma = \frac{\beta_b}{2\beta} \geq 0$.²⁷

- 2. Skill-price effect in encouraged firms** (*Encouraged non-adopters vs. non-encouraged non-adopters*)

$$\Delta_{\text{E,NA} - \text{NE,NA}}^Y = -\kappa/(\epsilon - 1). \quad (36)$$

- 3. Combined effects for encouraged adopters** (*Encouraged adopters vs. non-encouraged non-adopters*)

$$\Delta_{\text{E,A} - \text{NE,NA}}^Y = -\kappa/(\epsilon - 1) + \gamma(\alpha_{c0} + \alpha_{c1}) + (1 - \gamma)(\alpha_{b0} + \alpha_{b1}). \quad (37)$$

With perfect worker mobility, Equation (31) implies an additional $\gamma\kappa/(\epsilon - 1)$ term.

Workplace-level difference-in-differences. Section 3.4 estimates the effect of encouragement on workplace outcomes:

- 1. Employment effect:**

$$\Delta_{\text{E} - \text{NE}}^N = \frac{\epsilon - \gamma}{\epsilon - 1}\kappa - (1 - \gamma)(\alpha_{b0} + \alpha_{b1}) - \gamma(\alpha_{c0} + \alpha_{c1}). \quad (38)$$

- 2. Wage bill effect:**

$$\Delta_{\text{E} - \text{NE}}^{WH} = \kappa. \quad (39)$$

A.2.6 Missing Intercept

The estimators in Equations (35)–(37) identify earnings effects up to the skill price effect in non-encouraged firms, $d\log W_0$ —the familiar missing intercept problem, since within-sector

²⁷The frictionless sorting model predicts all adopters sort into encouraged firms, leaving no workers to identify Equation (35). In practice, Section 3 finds little worker resorting, so Equation (35) derives the structural interpretation without mobility.

difference-in-differences cannot capture sector-wide general equilibrium effects.

We use the model to characterize this residual. The key insight is that adopters generate spillovers to non-adopters only when their own market outcomes change; null effects on adopters' market outcomes thus imply limited spillovers.

To formalize, assume nested CES demand with elasticity η across and ϵ within sectors:

$$Y_j = P_j^{-\epsilon} P^{\epsilon-\eta}, \quad (40)$$

where $P = (\int P_j^{1-\epsilon} dj)^{\frac{1}{1-\epsilon}}$ is the sectoral price index. Combining with Equations (11), (14), (16), and (40):

$$d \log W_0 = \underbrace{-\frac{1}{\epsilon} d \log Y_0}_{\text{Residual demand}} + \underbrace{\frac{\eta - \epsilon}{\epsilon \eta} d \log Y}_{\text{Sector competition}}, \quad (41)$$

where the first term reflects firms moving down residual demand curves and the second reflects competing firms stealing sectoral demand.

The change in non-encouraged output reflects both productivity and employment:

$$d \log Y_0 = \underbrace{\mathbb{E}[A_{i0} B_{i0}]}_{\text{Adopter productivity}} + \underbrace{\mathbb{E}[d \log N_0]}_{\text{Employment}}, \quad (42)$$

where these terms are identified by:

$$\mathbb{E}[A_{i0} B_{i0}] = A_0 \times \Delta_{\text{NE}, \text{A-NE}, \text{NA}}^Y, \quad (43)$$

$$\mathbb{E}[d \log N_0] = -E \times \Delta_{\text{E-NE}}^N, \quad (44)$$

with A_0 the adoption rate in non-encouraged firms, E the share of encouraging firms, and Equation (44) using the labor market clearing condition. Similarly, total output growth reflects aggregate adopter productivity:

$$d \log Y = (1 - E) A_0 \cdot \Delta_{\text{NE}, \text{A-NE}, \text{NA}}^Y + E A_1 \cdot \Delta_{\text{E}, \text{A-E}, \text{NA}}^Y, \quad (45)$$